

Giacomo Medda

Resume

Last updated: July 3, 2026

Machine learning researcher specializing in recommender systems, working across generative, transformer, and graph-based models for personalization, cold-start, and explainability. Experience building and deploying end-to-end explainable systems, with a focus on translating research ideas into practical, scalable AI. Values environments where ideas are respected and growth is collective.

PROFESSIONAL EXPERIENCE

MAR 2026 — PRESENT (PT)

University of Cagliari (Computer Science Department)
*University Lecturer of **Structured Computer Organization***

FEB 2025 — PRESENT (FT)

University of Cagliari (Computer Science Department)
Research Collaborator

Researching **semantic IDs**, **generative recommendation**, and **LLM personas**. Developing **memory network** architectures for **multimodal cold-start**. Deployed a live user-study platform serving a **path-based transformer** for real-time explainable recommendations from cold-start interactions. Supervising student research toward publication.

2024 — 2026 (PT)

Various institutions (Univ. Cagliari, Sardegna Ricerche, Pacinotti High School)
University Tutor and Outreach Activities in AI

JAN 2024 — JAN 2025 (FT)

University of Cagliari (Computer Science Department)
Postdoctoral Researcher (PHaSE Project)

Built an explainable food recommender using **KG path reasoning** and **LLM-based data augmentation** for 300k+ sustainable food choices. Deployed a containerized system powering an LLM-driven chatbot.

OCT 2020 — FEB 2024 (FT)

University of Cagliari (Computer Science Department)
Ph.D. Student

Developed **graph-based** and **counterfactual** models with evaluation frameworks for **fair** and **explainable recommender** and **speaker recognition** systems, achieving state-of-the-art bias mitigation.

APR 2022 — JUL 2022 (FT)

Eurecat (Data Science and Big Data Analytics unit)
Research Intern

Collaborated on **explainable unfairness** analysis in **GNN-based recommenders**, leading the implementation of fair diagnostic tools.

FEB 2020 — JUL 2020 (FT)

Flosslab
External Consultant (Software Engineer)

Developed an automated presence detector for the university platform supporting remote learning during COVID-19.

EDUCATION

2020 — 2024 **PhD - Recommender Systems & Responsible AI**
Mathematics and Computer Science, University of Cagliari

2018 — 2020 **MSc Computer Science** (MAGNA CUM LAUDE)
University of Cagliari

2015 — 2018 **BSc Computer Science** (MAGNA CUM LAUDE)
University of Cagliari

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TECHNICAL SKILLS

Programming

Python (primary); *C++*, *Java*, *SQL* (familiar); *Rust* (in progress)

Machine Learning & Deep Learning

PyTorch, *PyG*, *scikit-learn*, *TensorFlow*

ML systems development, evaluation, and experimentation.

Data Processing & Analysis

Polars, *Pandas*, *NumPy*, *SciPy*, *Matplotlib*

EDA, statistical evaluation, and data visualization.

APIs & Experiment Infrastructure

FastAPI, *Docker*, *GitHub Actions*

Automated testing, containerization, docs deployment (CI/CD).

Language Models & Generative AI

HuggingFace Transformers, *Ollama*, *llama-cpp*

Explainable recommenders, data annotation and extraction.

Cloud & Big Data

AWS, *Terraform*, *PySpark*, *Hadoop* (**distributed systems**)

SELECTED PROJECTS

hopwise — Open-source recsys library extending RecBole with knowledge-graph path reasoning and explainable recommendation; supports training decoder-only transformers from scratch for autoregressive path generation (CIKM 2025 resource paper; UMAP 2026 tutorial).

PHaSE API — Containerized FastAPI service enabling zero-shot sustainable food recommendation from preference cues over 300k+ items, powering an LLM-driven chatbot.

FA4GCF — GNN recommendation baselines implemented from scratch on an extended RecBole, with graph data augmentation policies and a novel fairness loss (RecSys 2024).

ACADEMIC SERVICE

Conference & Workshop Organizer: ECIR, UMAP, HT

Program Committee & Reviewer: RecSys, CIKM, SIGIR, WSDM, WWW, ECIR, UMAP, ECML-PKDD, ECAI

Journal Guest Editor: Springer DMKD

SELECTED PUBLICATIONS

Note: In papers with alphabetical author order, I am the primary author.

- [1] **Medda, G.**, et al., “FoodNexus: Massive Food Knowledge for Recommender Systems.”, *ECIR* 2026.
- [2] **Medda, G.**, et al., “hopwise: A Python Library for Explainable Recommendation based on Path Reasoning over Knowledge Graphs”, *CIKM* 2025.
- [3] **Medda, G.**, et al., “GNNUERS: Fairness Explanation in GNNs for Recommendation via Counterfactual Reasoning.”, *ACM TIST* (2025).
- [4] Malitesta, D., **Medda, G.**, et al., “How Fair is Your Diffusion Recommender Model?”, *RecSys* 2025.
- [5] **Medda, G.**, et al., “Fair Augmentation for Graph Collaborative Filtering”, *RecSys* 2024.
- [6] **Medda, G.**, et al., “GreenFoodLens: Sustainability Labels for Food Recommendation”, *RecSys* 2025.